
TOPS TECH Task-2 (CS)

Task-2 Types of hackers

- 1. Create a table listing and describing the main types of hackers (such as white hat, black hat, grey hat, script kiddie, hacktivist, and state-sponsored) and give one real-world example or news headline for each type.**

Ans:

Types of Hackers Table

Type of Hacker	Description	Real-World Example / News Headline
White Hat	Ethical hackers who work with permission from organizations to identify and fix security vulnerabilities. Their goal is to improve cybersecurity and protect users.	Google's Project Zero team legally discovers and reports software vulnerabilities to vendors so they can be fixed.
Black Hat	Malicious hackers who gain unauthorized access to systems for personal gain, financial profit, data theft, or causing damage.	The REvil ransomware group carried out attacks against multiple organizations, encrypting data and demanding ransom payments.
Grey Hat	Hackers who may access systems without permission but usually do not intend to cause harm. They often disclose vulnerabilities after discovering them.	In 2013, a security researcher demonstrated a Facebook vulnerability by posting on Mark Zuckerberg's timeline after previous reports were ignored.
Script Kiddie	Beginners or inexperienced individuals who use pre-built hacking tools created by others rather than developing their own techniques.	Some attackers use tools such as LOIC (Low Orbit Ion Cannon) to launch DDoS attacks without advanced technical knowledge.

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Type of Hacker	Description	Real-World Example / News Headline
Hacktivist	Hackers who conduct cyberattacks to promote political, social, or ideological causes. Their actions are often intended to raise awareness or protest against organizations.	The Anonymous collective has conducted cyber campaigns against governments and organizations to support various causes.
State-Sponsored	Hackers who are funded or supported by governments to perform cyber espionage, intelligence gathering, or cyber warfare operations.	The Stuxnet cyber operation targeted Iran's nuclear facilities and is widely considered an example of a state-sponsored cyberattack.

⇒ Hackers can be classified based on their intentions and activities. White Hat hackers help improve security, Black Hat hackers engage in illegal activities, Grey Hat hackers operate between the two, Script Kiddies rely on existing tools, Hacktivists support causes through cyber actions, and State-Sponsored hackers work on behalf of governments.

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2. Pick any popular app you use (Instagram, Paytm, Zomato, etc.) and describe a scenario for each type of hacker: how would a white hat, black hat, and grey hat hacker interact with this app? Write 2-3 sentences per scenario.

Ans.

Hacker Scenarios Using Instagram

White Hat Hacker

- A White Hat hacker works with Instagram's permission to identify security vulnerabilities.
- They test the platform in a legal and ethical manner and report any discovered issues through responsible disclosure or bug bounty programs.
- Their goal is to help Instagram improve security and protect user data.

Black Hat Hacker

- A Black Hat hacker attempts to gain unauthorized access to Instagram accounts for personal gain.
- They may use phishing attacks, credential theft, or malware to steal usernames, passwords, and personal information.
- Their actions are illegal and can cause financial or reputational damage to users.

Grey Hat Hacker

- A Grey Hat hacker may test Instagram for vulnerabilities without obtaining prior permission.
- Although they usually do not intend to cause harm, their actions still violate the platform's rules.
- After discovering a security weakness, they may inform Instagram and suggest fixes.

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Script Kiddie

- A Script Kiddie uses publicly available hacking tools and scripts to target Instagram accounts without fully understanding how the tools work.
- They often attempt password attacks or other basic exploits using software created by experienced hackers.
- Their technical knowledge is usually limited.

Hacktivist

- A Hacktivist may target Instagram accounts or pages to promote a political, social, or ideological cause.
- They might deface content, spread messages, or disrupt services to attract public attention.
- Their primary motivation is activism rather than financial gain.

State-Sponsored Hacker

- A State-Sponsored hacker may target Instagram to collect intelligence, monitor specific individuals, or conduct cyber operations on behalf of a government.
- These hackers are highly skilled and often have significant resources and support.
- Their activities are usually related to national security or strategic objectives.

⇒ Different types of hackers interact with the same application in different ways. White Hat hackers help improve security, Black Hat hackers exploit weaknesses for illegal purposes, Grey Hat hackers operate in a legally questionable area, Script Kiddies rely on existing tools, Hacktivists promote causes, and State-Sponsored hackers serve government interests.

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3. Given the following user actions, classify each as white hat, black hat, grey hat, or other: reporting a bug to the company, stealing user data, testing your own account for vulnerabilities, defacing a website for a political cause, and participating in a government cyber operation.
Hint: Use your table from Task 1 for quick reference.

Ans.

Classification of User Actions

User Action	Classification	Explanation
Reporting a bug to the company	White Hat	Reporting a security vulnerability responsibly helps the organization fix the issue and improve security.
Stealing user data	Black Hat	Unauthorized access to personal information is illegal and is a common activity of Black Hat hackers.
Testing your own account for vulnerabilities	White Hat / Ethical Activity	Testing the security of your own account is generally considered ethical because you have permission to access it.
Defacing a website for a political cause	Hacktivist	Website defacement to promote political or social messages is a common form of hacktivism.

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User Action	Classification	Explanation
Participating in a government cyber operation	State-Sponsored	Cyber operations conducted on behalf of a government are classified as state-sponsored activities.

⇒ The classification of an action depends on the hacker's intention, authorization, and purpose. Ethical actions such as reporting bugs are associated with White Hat hackers, while illegal activities such as data theft are linked to Black Hat hackers. Political cyber activities are often connected to Hacktivists, and government-backed operations are classified as State-Sponsored.

4. Interview-style: Imagine you are explaining to your non-technical friend the difference between a white hat and a black hat hacker using a cricket match analogy. Write your explanation in 4-5 sentences.

Ans:

White Hat vs Black Hat Hacker (Cricket Match Analogy)

- Imagine a cricket match where following the rules is important for fair play.
- A White Hat hacker is like an umpire who follows the rules and helps keep the game fair and secure.
- A Black Hat hacker is like a player who cheats by breaking the rules to gain an unfair advantage.
- White Hat hackers use their skills to find and report security vulnerabilities so they can be fixed.
- Black Hat hackers exploit vulnerabilities to steal data, earn money illegally, or cause damage to systems.
- Although both have technical skills, their intentions and actions are completely different.